

The total number of poles (counted according to their multiplicities) of an elliptic function is called its **order**. The next theorem says that elliptic functions have as many zeros as they have poles, if the zeros are counted with their multiplicities.

Theorem 1.4 *Every elliptic function of order m has m zeros in P_0 .*

Proof. Assuming first that f has no zeros or poles on the boundary of P_0 , we know by the argument principle in Chapter 3 that

$$\int_{\partial P_0} \frac{f'(z)}{f(z)} dz = 2\pi i(\mathcal{N}_3 - \mathcal{N}_p)$$

where $\mathcal{N}_3$ and $\mathcal{N}_p$ denote the number of zeros and poles of f in P_0 , respectively. By periodicity, we can argue as in the proof of the previous theorem to find that $\int_{\partial P_0} f'/f = 0$, and therefore $\mathcal{N}_3 = \mathcal{N}_p$.

In the case when a pole or zero of f lies on ∂P_0 it suffices to apply the argument to a translate of P .

As a consequence, if f is elliptic then the equation $f(z) = c$ has as many solutions as the order of f for every $c \in \mathbb{C}$, simply because $f - c$ is elliptic and has as many poles as f .

Despite the rather simple nature of the theorems above, there remains the question of showing that elliptic functions exist. We now turn to a constructive solution of this problem.

1.2 The Weierstrass $\wp$ function

An elliptic function of order two

This section is devoted to the basic example of an elliptic function. As we have seen above, any elliptic function must have at least two poles; we shall in fact construct one whose only singularity will be a double pole at the points of the lattice generated by the periods.

Before looking at the case of doubly-periodic functions, let us first consider briefly functions with only a single period. If one wished to construct a function with period 1 and poles at all the integers, a simple choice would be the sum

$$F(z) = \sum_{n=-\infty}^{\infty} \frac{1}{z+n}.$$

Note that the sum remains unchanged if we replace z by $z + 1$, and the poles are at the integers. However, the series defining F is not absolutely

convergent, and to remedy this problem, we sum symmetrically, that is, we define

$$F(z) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \left[\frac{1}{z+n} + \frac{1}{z-n} \right].$$

On the far right-hand side, we have paired up the terms corresponding to n and $-n$, a trick which makes the quantity in brackets $O(1/n^2)$, and hence the last sum is absolutely convergent. As a consequence, F is meromorphic with poles precisely at the integers. In fact, we proved earlier in Chapter 5 that $F(z) = \pi \cot \pi z$.

There is a second way to deal with the series $\sum_{-\infty}^{\infty} 1/(z+n)$, which is to write it as

$$\frac{1}{z} + \sum_{n \neq 0} \left[\frac{1}{z+n} - \frac{1}{n} \right],$$

where the sum is taken over all non-zero integers. Notice that $1/(z+n) - 1/n = O(1/n^2)$, which makes this series absolutely convergent. Moreover, since

$$\frac{1}{z+n} + \frac{1}{z-n} = \left(\frac{1}{z+n} - \frac{1}{n} \right) + \left(\frac{1}{z-n} - \frac{1}{-n} \right),$$

we get the same sum as before.

In analogy to this, the idea is to mimic the above to produce our first example of an elliptic function. We would like to write it as

$$\sum_{\omega \in \Lambda} \frac{1}{(z+\omega)^2},$$

but again this series does not converge absolutely. There are several approaches to try to make sense of this series (see Problem 1), but the simplest is to follow the second way we dealt with the cotangent series.

To overcome the non-absolute convergence of the series, let Λ^* denote the lattice minus the origin, that is, $\Lambda^* = \Lambda - \{(0,0)\}$, and consider instead the following series:

$$\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right],$$

where we have subtracted the factor $1/\omega^2$ to make the sum converge. The term in brackets is now

$$\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2 - 2z\omega}{(z+\omega)^2\omega^2} = O\left(\frac{1}{\omega^3}\right) \quad \text{as } |\omega| \rightarrow \infty,$$

and the new series will define a meromorphic function with the desired poles once we have proved the following lemma.

Lemma 1.5 *The two series*

$$\sum_{(n,m) \neq (0,0)} \frac{1}{(|n| + |m|)^r} \quad \text{and} \quad \sum_{n+m\tau \in \Lambda^*} \frac{1}{|n + m\tau|^r}$$

converge if $r > 2$.

Recall that according to the Note at the end of Chapter 7, the question whether a double series converges absolutely is independent of the order of summation. In the present case, we shall first sum in m and then in n .

For the first series, the usual integral comparison can be applied.² For each $n \neq 0$

$$\begin{aligned} \sum_{m \in \mathbb{Z}} \frac{1}{(|n| + |m|)^r} &= \frac{1}{|n|^r} + 2 \sum_{m \geq 1} \frac{1}{(|n| + |m|)^r} \\ &= \frac{1}{|n|^r} + 2 \sum_{k \geq |n|+1} \frac{1}{k^r} \\ &\leq \frac{1}{|n|^r} + 2 \int_{|n|}^{\infty} \frac{dx}{x^r} \\ &\leq \frac{1}{|n|^r} + C \frac{1}{|n|^{r-1}}. \end{aligned}$$

Therefore, $r > 2$ implies

$$\begin{aligned} \sum_{(n,m) \neq (0,0)} \frac{1}{(|n| + |m|)^r} &= \sum_{|m| \neq 0} \frac{1}{|m|^r} + \sum_{|n| \neq 0} \sum_{m \in \mathbb{Z}} \frac{1}{(|n| + |m|)^r} \\ &\leq \sum_{|m| \neq 0} \frac{1}{|m|^r} + \sum_{|n| \neq 0} \left(\frac{1}{|n|^r} + C \frac{1}{|n|^{r-1}} \right) \\ &< \infty. \end{aligned}$$

To prove that the second series also converges, it suffices to show that there is a constant c such that

$$|n| + |m| \leq c|n + \tau m| \quad \text{for all } n, m \in \mathbb{Z}.$$

²We simply use $1/k^r \leq 1/x^r$ when $k-1 \leq x \leq k$; see also the first figure in Chapter 8, Book I.

We use the notation $x \lesssim y$ if there exists a positive constant a such that $x \leq ay$. We also write $x \approx y$ if both $x \lesssim y$ and $y \lesssim x$ hold. Note that for any two positive numbers A and B , one has

$$(A^2 + B^2)^{1/2} \approx A + B.$$

On the one hand $A \leq (A^2 + B^2)^{1/2}$ and $B \leq (A^2 + B^2)^{1/2}$, so that $A + B \leq 2(A^2 + B^2)^{1/2}$. On the other hand, it suffices to square both sides to see that $(A^2 + B^2)^{1/2} \leq A + B$.

The proof that the second series in Lemma 1.5 converges is now a consequence of the following observation:

$$|n| + |m| \approx |n + m\tau| \quad \text{whenever } \tau \in \mathbb{H}.$$

Indeed, if $\tau = s + it$ with $s, t \in \mathbb{R}$ and $t > 0$, then

$$|n + m\tau| = [(n + ms)^2 + (mt)^2]^{1/2} \approx |n + ms| + |mt| \approx |n + ms| + |m|,$$

by the previous observation. Then, $|n + ms| + |m| \approx |n| + |m|$, by considering separately the cases when $|n| \leq 2|m||s|$ and $|n| \geq 2|m||s|$.

Remark. The proof above shows that when $r > 2$ the series $\sum |n + m\tau|^{-r}$ converges uniformly in every half-plane $\text{Im}(\tau) \geq \delta > 0$.

In contrast, when $r = 2$ this series fails to converge (Exercise 3).

With this technical point behind us, we may now return to the definition of the **Weierstrass $\wp$ function**, which is given by the series

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] \\ &= \frac{1}{z^2} + \sum_{(n,m) \neq (0,0)} \left[\frac{1}{(z + n + m\tau)^2} - \frac{1}{(n + m\tau)^2} \right]. \end{aligned}$$

We claim that $\wp$ is a meromorphic function with double poles at the lattice points. To see this, suppose that $|z| < R$, and write

$$\wp(z) = \frac{1}{z^2} + \sum_{|\omega| \leq 2R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] + \sum_{|\omega| > 2R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right].$$

The term in the second sum is $O(1/|\omega|^3)$ uniformly for $|z| < R$, so by Lemma 1.5 this second sum defines a holomorphic function in $|z| < R$. Finally, note that the first sum exhibits double poles at the lattice points in the disc $|z| < R$.

Observe that because of the insertion of the terms $-1/\omega^2$, it is no longer obvious whether $\wp$ is doubly periodic. Nevertheless this is true, and $\wp$ has all the properties of an elliptic function of order 2. We gather this result in a theorem.

Theorem 1.6 *The function $\wp$ is an elliptic function that has periods 1 and τ , and double poles at the lattice points.*

Proof. It remains only to prove that $\wp$ is periodic with the correct periods. To do so, note that the derivative is given by differentiating the series for $\wp$ termwise so

$$\wp'(z) = -2 \sum_{n,m \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^3}.$$

This accomplishes two things for us. First, the differentiated series converges absolutely whenever z is not a lattice point, by the case $r = 3$ of Lemma 1.5. Second, the differentiation also eliminates the subtraction term $1/\omega^2$; therefore the series for $\wp'$ is clearly periodic with periods 1 and τ , since it remains unchanged after replacing z by $z + 1$ or $z + \tau$.

Hence, there are two constants a and b such that

$$\wp(z + 1) = \wp(z) + a \quad \text{and} \quad \wp(z + \tau) = \wp(z) + b.$$

It is clear from the definition, however, that $\wp$ is even, that is, $\wp(z) = \wp(-z)$, since the sum over $\omega \in \Lambda$ can be replaced by the sum over $-\omega \in \Lambda$. Therefore $\wp(-1/2) = \wp(1/2)$ and $\wp(-\tau/2) = \wp(\tau/2)$, and setting $z = -1/2$ and $z = -\tau/2$, respectively, in the two expressions above proves that $a = b = 0$.

A direct proof of the periodicity of $\wp$ can be given without differentiation; see Exercise 4.

Properties of $\wp$

Several remarks are in order. First, we have already observed that $\wp$ is even, and therefore $\wp'$ is odd. Since $\wp'$ is also periodic with periods 1 and τ , we find that

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\left(\frac{1+\tau}{2}\right) = 0.$$

Indeed, one has, for example,

$$\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2 + 1) = -\wp'(1/2).$$

Since $\wp'$ is elliptic and has order 3, the three points $1/2$, $\tau/2$, and $(1+\tau)/2$ (which are called the **half-periods**) are the only roots of $\wp'$ in the fundamental parallelogram, and they have multiplicity 1. Therefore, if we define

$$\wp(1/2) = e_1, \quad \wp(\tau/2) = e_2, \quad \text{and} \quad \wp\left(\frac{1+\tau}{2}\right) = e_3,$$

we conclude that the equation $\wp(z) = e_1$ has a double root at $1/2$. Since $\wp$ has order 2, there are no other solutions to the equation $\wp(z) = e_1$ in the fundamental parallelogram. Similarly the equations $\wp(z) = e_2$ and $\wp(z) = e_3$ have only double roots at $\tau/2$ and $(1+\tau)/2$, respectively. In particular, the three numbers e_1, e_2 , and e_3 are distinct, for otherwise $\wp$ would have at least four roots in the fundamental parallelogram, contradicting the fact that $\wp$ has order 2. From these observations we can prove the following theorem.

Theorem 1.7 *The function $(\wp')^2$ is the cubic polynomial in $\wp$*

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. The only roots of $F(z) = (\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3)$ in the fundamental parallelogram have multiplicity 2 and are at the points $1/2, \tau/2$, and $(1+\tau)/2$. Also, $(\wp')^2$ has double roots at these points. Moreover, F has poles of order 6 at the lattice points, and so does $(\wp')^2$ (because $\wp'$ has poles of order 3 there). Consequently $(\wp')^2/F$ is holomorphic and still doubly-periodic, hence this quotient is constant. To find the value of this constant we note that for z near 0, one has

$$\wp(z) = \frac{1}{z^2} + \cdots \quad \text{and} \quad \wp'(z) = \frac{-2}{z^3} + \cdots,$$

where the dots indicate terms of higher order. Therefore the constant is 4, and the theorem is proved.

We next demonstrate the universality of $\wp$ by showing that every elliptic function is a simple combination of $\wp$ and $\wp'$.

Theorem 1.8 *Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.*

The theorem will be an easy consequence of the following version of it.

Lemma 1.9 *Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.*

Proof. If F has a zero or pole at the origin it must be of even order, since F is an even function. As a consequence, there exists an integer m so that $F\wp^m$ has no zero or pole at the lattice points. We may therefore assume that F itself has no zero or pole on Λ .

Our immediate goal is to use $\wp$ to construct a doubly-periodic function G with precisely the same zeros and poles as F . To achieve this, we recall that $\wp(z) - \wp(a)$ has a single zero of order 2 if a is a half-period, and two distinct zeros at a and $-a$ otherwise. We must therefore carefully count the zeros and poles of F .

If a is a zero of F , then so is $-a$, since F is even. Moreover, a is congruent to $-a$ if and only if it is a half-period, in which case the zero is of even order. Therefore, if the points $a_1, -a_1, \dots, a_m, -a_m$ counted with multiplicities³ describe all the zeros of F , then

$$[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]$$

has precisely the same roots as F . A similar argument, where $b_1, -b_1, \dots, b_m, -b_m$ (with multiplicities) describe all the poles of F , then shows that

$$G(z) = \frac{[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)] \cdots [\wp(z) - \wp(b_m)]}$$

is periodic and has the same zeros and poles as F . Therefore, F/G is holomorphic and doubly-periodic, hence constant. This concludes the proof of the lemma.

To prove the theorem, we first recall that $\wp$ is even while $\wp'$ odd. We then write f as a sum of an even and an odd function,

$$f(z) = f_{\text{even}}(z) + f_{\text{odd}}(z),$$

where in fact

$$f_{\text{even}}(z) = \frac{f(z) + f(-z)}{2} \quad \text{and} \quad f_{\text{odd}}(z) = \frac{f(z) - f(-z)}{2}.$$

Then, since $f_{\text{odd}}/\wp'$ is even, it is clear from the lemma applied to f_{even} and $f_{\text{odd}}/\wp'$ that f is a rational function of $\wp$ and $\wp'$.

³If a_j is not a half-period, then a_j and $-a_j$ have the multiplicity of F at these points. If a_j is a half-period, then a_j and $-a_j$ are congruent and each has multiplicity half of the multiplicity of F at this point.

2 The modular character of elliptic functions and Eisenstein series

We shall now study the modular character of elliptic functions, that is, their dependence on τ .

Recall the normalization we made at the beginning of the chapter. We started with two periods ω_1 and ω_2 linearly that are independent over $\mathbb{R}$, and we defined $\tau = \omega_2/\omega_1$. We could then assume that $\text{Im}(\tau) > 0$, and also that the two periods are 1 and τ . Next, we considered the lattice generated by 1 and τ and constructed the function $\wp$, which is elliptic of order 2 with periods 1 and τ . Since the construction of $\wp$ depends on τ , we could write $\wp_\tau$ instead. This leads us to change our point of view and think of $\wp_\tau(z)$ primarily as a function of τ . This approach yields many interesting new insights.

Our considerations are guided by the following observations. First, since 1 and τ generate the periods of $\wp_\tau(z)$, and 1 and $\tau + 1$ generate the same periods, we can expect a close relationship between $\wp_\tau(z)$ and $\wp_{\tau+1}(z)$. In fact, it is easy to see that they are identical. Second, since $\tau = \omega_2/\omega_1$, by the normalization imposed at the beginning of Section 1, we see that $-1/\tau = -\omega_1/\omega_2$ (with $\text{Im}(-1/\tau) > 0$). This corresponds essentially to an interchange of the two periods ω_1 and ω_2 , and thus we can also expect an intimate connection between $\wp_\tau$ and $\wp_{-1/\tau}$. In fact, it is easy to verify that $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$.

So we are led to consider the group of transformations of the upper half-plane $\text{Im}(\tau) > 0$, generated by the two transformations $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. This group is called the **modular group**. On the basis of what we said, it can be expected that all quantities intrinsically attached to $\wp_\tau(z)$ reflect the above transformations. We see this clearly when we consider the Eisenstein series.

2.1 Eisenstein series

The **Eisenstein series** of order k is defined by

$$E_k(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n + m\tau)^k},$$

whenever k is an integer ≥ 3 and τ is a complex number with $\text{Im}(\tau) > 0$. If Λ is the lattice generated by 1 and τ , and if we write $\omega = n + m\tau$, then another expression for the Eisenstein series is $\sum_{\omega \in \Lambda^*} 1/\omega^k$.

Theorem 2.1 *Eisenstein series have the following properties:*